



World Logics

LoDE Decision Procedures

(HP2T)

World Logics Decision Procedures

- **Introduction**
- Unfolding
- LoDE language and knowledge reasoning
- Expansion
- LoDE entity reasoning
- Key notions

LoDE – Language and knowledge reasoning problems (Recall)

Definition (LoDE language and knowledge reasoning problem). LoDE language and knowledge reasoning problems are reasoning problems $T \models_{\{M\}} C$ where both $M \models T$ and $M \models C$ are solved in LoD (LoC).

Observation (LoDE language and knowledge reasoning problem). LoDE language and knowledge reasoning problems apply to the KG theories of the following base and composed world models:

- LoC: Concept graphs (CG)
- LoD: Etype (entity type) graphs (ETG)
- LoCD (LoC+LoD): Concept + etype graphs (CETG)

LoDE – entity reasoning problems (Recall)

Definition (LoDE Entity Reasoning problem). LoDE entity reasoning problems are reasoning problems $T \models_{\{M\}} C$ where $M \models T$ is solved in LoD (LoC) while $M \models C$ is solved in LoE.

Observation (LoDE entity reasoning problem). LoDE entity reasoning problems apply to the KG theories of the following base and composed world models

- LoE: Entity graphs (EG)
- LoCE (LoC + LoE): Concept + entity graphs (EG)
- LoDE (LoD + LoE): Etype + entity graphs (EETG)
- LoCDE (LoC+LoD+LoE): Concept + etype + entity graphs (CEETG)

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LoD defined and primitive etype

Definition (Definiendum, definiens): The left hand side of a definition $A \equiv C$ (that is, A) is called **definiendum**, the right hand side (that is, C) is called the definiens.

Definition (Defined and primitive etype): Given a TBox, a **defined** etype is an etype which appears on the left of a definition of the TBox. A **primitive** etype is an etype which only appears on the right of the definitions. A primitive type is an **atomic** etype. A defined etype is a **complex** etype.

Reminder (Primitive etypes). Primitive etypes are etypes defined from LoC concepts, i.e., etypes which are equivalent to LoC concepts.

Definition (LoC-to-LoD definition) If C is a **LoC concept** and E is a LoD primitive etype we have

$$E \equiv C$$

as a so called **LoC-to-LoD definition**. LoC-to-LoD definitions are parts of LoD KG theories.

Unfolding an etype

Definition (etype unfolding) A defined etype is **unfolded** if all the defined etypes occurring in its definiens are recursively substituted with their definition.

Observation (unfolded etype): The definition of an unfolded etype contains only primitive etypes

Example (unflded etype).

From:

$\text{ElectricGuitar} \equiv \text{Guitar} \sqcap \text{SoundAmplification}$

$\text{ElectricGuitar\#1} \equiv \text{ElectricGuitar} \sqcap \exists \text{hasColour.String} \sqcap \exists \text{hasBrand.String}$

... to:

$\text{ElectricGuitar\#1} \equiv \text{Guitar} \sqcap \text{SoundAmplification} \sqcap \exists \text{hasColour.String} \sqcap \exists \text{hasBrand.String}$

Remark: In an acyclic terminology the process of etype unfolding is applied recursively up to any level, usually with the goal of reducing to primitive etypes. The process is guaranteed to terminate.

Unfolding a TBox

Observation (Defined and primitive etype): We restrict to the case where a defined etype appears once on the left or a definition. Defined and primitive etypes can appear on the right of definitions any number of times.

Observation (TBox unfolding). A definitional TBox T can be unfolded into a Tbox T' which contains only primitive etypes by (recursively) unfolding all its defined etypes and by adding them to T' .

Definition (Unfolded TBox). A definitional TBox T unfolded contains only primitive etypes

Theorem (Unfolded Tbox) Let T be a LoD terminology. Let T' the result of unfolding T . Then M is a model of T if and only if it is a model of T' .

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Entailment with teleontologies

Theorem (Satisfiability): $T \models p$, that is, p satisfiable by T , iff, after unfolding there is no conjunct which occurs both negated and not negated in p

Theorem (Subsumption): $T \models p_1 \sqsubseteq p_2$ iff, after unfolding, the conjuncts in p_1 are a superset or equal to those of p_2

Theorem (Equivalence): $T \models p_1 \equiv p_2$ iff, after unfolding, the conjuncts in p_1 and in p_2 are exactly the same as those of p_2

Theorem (Disjointness): $T \models p_1 \perp p_2$ (the same as: $T \models p_1 \sqsubseteq \neg p_2$) iff, after unfolding, one of the conjuncts occurs negated in p_1 and not negated in p_2 , or vice versa

Entailment with teleontologies

Observation (reprise): Since we reason with teleontologies we are dealing with TBoxes with only conjuncts

Observation (Entailment with teleontologies): Teleontologies are nested subsumption hierarchies. Teleologies are unfolded teleontologies. That is, entailment in teleontologies is performed by first reducing teleontologies into teleologies.

Observation (Theorem 3): For $p_1 \sqsubseteq p_2$ to hold p_1 must have more (and not less!) conjuncts than p_2 . In fact, adding conjuncts makes a set smaller

Satisfiability with teleontologies (reprise)

- Bachelor $\equiv \text{Student} \sqcap \text{Undergraduate}$
- Master $\equiv \text{Student} \sqcap \neg \text{Undergraduate}$
- PhD $\equiv \text{Student} \sqcap \neg \text{Undergraduate} \sqcap \text{Research}$
- Assistant $\equiv \text{Student} \sqcap \neg \text{Undergraduate} \sqcap \text{Research} \sqcap \text{Teach}$

Bachelor $\sqcap$ PhD satisfiable?

NO!

Observation. Assume (here and below) that all primitive etypes are inherited from LoC KG theories. Unfold the query using the conjuncts in T .

Subsumption with teleontologies (reprise)

- Bachelor $\equiv \text{Student} \sqcap \text{Undergraduate}$
- Master $\equiv \text{Student} \sqcap \neg \text{Undergraduate}$
- PhD $\equiv \text{Student} \sqcap \neg \text{Undergraduate} \sqcap \text{Research}$
- Assistant $\equiv \text{Student} \sqcap \neg \text{Undergraduate} \sqcap \text{Research} \sqcap \text{Teach}$

$T \models \text{PhD} \sqsubseteq \text{Student} ?$ YES!

Observation. Unfold the query using the conjuncts in T.

Equivalence with teleontologies (reprise)

- Bachelor $\equiv \text{Student} \sqcap \text{Undergraduate}$
- Master $\equiv \text{Student} \sqcap \neg \text{Undergraduate}$
- PhD $\equiv \text{Student} \sqcap \neg \text{Undergraduate} \sqcap \text{Research}$
- Assistant $\equiv \text{Student} \sqcap \neg \text{Undergraduate} \sqcap \text{Research} \sqcap \text{Teach}$

$T \models \text{Student} \equiv \text{Bachelor} \sqcup \text{Master}?$

YES!

Observation. Unfold the query using the conjuncts in T . Notice that we have extended the query language to allow for disjunction.

Disjointness with teleontologies (reprise)

- Bachelor $\equiv \text{Student} \sqcap \text{Undergraduate}$
- Master $\equiv \text{Student} \sqcap \neg \text{Undergraduate}$
- PhD $\equiv \text{Student} \sqcap \neg \text{Undergraduate} \sqcap \text{Research}$
- Assistant Teach $\equiv \text{Student} \sqcap \neg \text{Undergraduate} \sqcap \text{Research} \sqcap$

$T \models \text{Assistant} \perp \text{Undergraduate}?$

YES!

Observation. You could also check $\text{Assistant} \sqsubseteq \neg \text{Undergraduate}$

Complexity of TBox unfolding

Observation (TBox unfolding). TBox definitions are like macros that can be unfolded into primitive etypes.

Observation (Complexity of unfolding). The size of the unfolded TBox grows **exponentially** with the depth of the TBox induced subsumption hierarchy. For instance, from

$$A0 \equiv \exists r.A1 \sqcap \forall s.A1$$

$$A1 \equiv \exists r.A2 \sqcap \forall s.A2$$

$$A2 \equiv \exists r.A3 \sqcap \forall s.A3$$

... we obtain

$$A1 \equiv \exists r.(\exists r.A2 \sqcap \forall r.A2) \sqcap \forall s.(\exists r.A2 \sqcap \forall r.A2) \quad (A1 \text{ 2 times } A0)$$

$$A2 \equiv \exists r.(\exists r.(\exists r.A3 \sqcap \forall s.A3) \sqcap \forall s.(\exists r.A3 \sqcap \forall s.A3)) \sqcap \forall s.(\exists r.(\exists r.A3 \sqcap \forall s.A3) \sqcap \forall s.(\exists r.A3 \sqcap \forall s.A3)) \quad (A2 \text{ 4 times } A0)$$

Observation (Complexity of unfolding). The decision problem remains linear in the size of the unfolded Tbox. This would not be the case if we had disjunctions instead of conjunctions.

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Etype expansion

Definition (Etype expansion) Let T be an **unfolded** LoDE definitional TBox and A a LoDE ABox. Let C be an etype defined in T (after unfolding) as

$$C \equiv C_1 \sqcap \dots \sqcap C_n$$

where $C_1, \dots, C_n$ are conjuncts containing only primitive etypes.

Let e be an entity occurring in A . Assume $C(e)$ occurs in A . Then the **expansion** $\text{exp}(C(e))$ of $C(e)$ in A in T is defined as follows:

$$\text{exp}(C(e)) = \{C(e), \text{exp}(C_1(e)), \dots, \text{exp}(C_n(e))\}$$

Observation (expansion). Expansion is defined recursively. However the process is guaranteed to terminate because the TBox is acyclic by hypothesis (see lectures on KG theories).

Etype expansion (continued)

Definition (Etype expansion $\exp(C_i(e))$) Let $C_i(e)$, be an assertion in A which is a **conjunct** in the expansion of C . Then the **expansion $\exp(C_i(e))$** of the conjunct $C_i(e)$ is defined as follows

- If the conjunct C_i is a (primitive) etype C . Then

$$\exp(C(e)) = C(e)$$

- If the conjunct C_i is of the form $\exists P.C$. Then

$$\exp(\exists P.C(e)) = \{P(e, e_2), C(e_2)\}$$

with e_2 new in A .

- If the conjunct C_i is of the form $\forall P.C$, with $P(e, e_2)$. Then

$$\exp(\forall P.C(e)) = C(e_2)$$

Etype expansion (observations)

Observation (etype expansion). The expansion of an unfolded defined etype applied to an individual allows to expand it into multiple independent assertions, one per conjunct.

Observation (conjunct expansion). The expansion of conjunct consisting of an etype applied to an individual is the etype itself.

Observation (existential conjunct expansion). The expansion of conjunct consisting of an existential quantification allows to create a new link to an individual which is anonymous. For instance, the expansion of $\exists \text{HasChild.Person}(e1)$ is $\text{HasChild}(e1, \text{anonymous\#1})$ with $\text{anonymous\#1}$ not occurring in the EG .

Observation (universal conjunct expansion). The expansion of conjunct consisting of a universal quantification allows to know the etype of the individual in the codomain of the of the property. For instance, the expansio of $\forall \text{HasChild.Person}(e1)$ is $\text{Person}(e2)$ assuming that the link $\text{HasChild}(e1, e2)$ occurs in the EG.

Etype expansion (observations)

Observation (Size of an expansion). Let C be an etype defined in T (after unfolding) as

$$C \equiv C_1 \sqcap \dots \sqcap C_n$$

Then the expansion $\exp(C(e))$ of $C(e)$ in A in T is defined as follows:

$$\exp(C(e)) = \{C(e), \exp(C_1(e)), \dots, \exp(C_n(e))\},$$

for each entity e of etype C .

Observation (Complexity of unfolding, ETG vs EG). EGs are usually exponentially bigger than their reference ETGs, as a function of the number of entities in the EG.

Etype expansion (observations)

Observation (Scope of an expansion). A LoDE EG may contain both primitive and defined etypes. The choice of the defined etypes to expand is application and query dependent.

Observation (ETG granularity). For each assertion in the EG, the level of detail at which the etypes of an EG is expressed can be increased by expansion or decreased by its inverse operation, thus allowing for the fine tuning of the EG to the user questions.

Intuition (From LoDE entity reasoning to LoE reasoning). In LoDE, because of defined etypes, the same question can be posed at different levels of abstraction (e.g., vehicle vs. car vs. sportcar vs. Ferrari). Unfolding allows to tune the shape of the knowledge graph to the question. Expansion allows to build the proper EG for it where the etypes and properties of the query are aligned with those of the EG. At this point LoDE entity reasoning reduces to LoE reasoning).

Expansion of an unfolded TBox (example 3 (cont))

T=

- Bachelor $\equiv \text{Student} \sqcap \text{Undergrad}$
- Master $\equiv \text{Student} \sqcap \neg \text{Undergrad}$
- PhD $\equiv \text{Student} \sqcap \neg \text{Undergrad} \sqcap \text{Research}$
- Assistant $\equiv \text{Student} \sqcap \neg \text{Undergrad} \sqcap \text{Research} \sqcap \text{Teach}$

A = {PhD(Rui)}

C = PhD

$\text{Exp}(\text{PhD}(\text{Rui})) = \{\text{PhD}(\text{Rui}), \text{Student}(\text{Rui}), \neg \text{Undergrad}(\text{Rui}), \text{Research}(\text{Rui}), \text{Master}(\text{Rui})\}$

Expansion - example

$T = \{ \text{Mother} \equiv \text{Female} \sqcap \exists \text{HasChild}.\text{Person}, \text{Female} \equiv \text{Person} \sqcap \text{Nice} \}$

$A = \{ \text{Mother}(\text{Anna}) \}$

$C = \text{Mother}$

The expansion A' of A with respect to T is:

$A' = \{ \text{Mother}(\text{Anna}), \text{Female}(\text{Anna}), \text{HasChild}(\text{Anna}, \text{Anonymous\#1}), \text{Person}(\text{Anna}), \text{Person}(\text{Anonymous\#1}), \text{Nice}(\text{Anna}) \}$

Expansion - example

$T = \{ \text{FemaleMother} \equiv \text{Female} \sqcap \forall \text{HasChild.female}, \text{Female} \equiv \text{Person} \sqcap \text{Nice} \}$

$A = \{ \text{FemaleMother}(\text{Anna}), \text{HasChild}(\text{Anna}, \text{Mary}) \}$

$C = \text{FemaleMother}$

The expansion A' of A with respect to T is:

$A' = \{ \text{FemaleMother}(\text{Anna}), \text{HasChild}(\text{Anna}, \text{Mary}), \text{Female}(\text{Anna}), \text{Female}(\text{Mary}), \text{Person}(\text{Anna}), \text{Nice}(\text{Anna}), \text{Person}(\text{Mary}), \text{Nice}(\text{Mary}) \}$

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Expansion (main result)

Definition (ABox expansion wrt a TBox). Apply etype expansion to all occurrences of all etypes occurring in the ABox and defined in the terminology T' obtained by unfolding the TBox T . The ABox A' obtained by adding A all its etype expansions is called the **expansion** of A with respect to T .

Theorem. Let T be a terminology and A and an ABox. Let A' be the result of expanding A based on the terminology T' obtained by unfolding T . Then M is a model of $T \cup A$ if and only if M is a model of A' .

$$T \models_M A \text{ if and only if } M \models T \cup A \text{ if and only if } M \models A'$$

LoDE entity reasoning

Theorem (LoDE entity reasoning). Let A' be an ABox expansion of the ABox A based on the terminology T' obtained by unfolding the TBox T . Let $a' \in \exp(a)$, with $a \in A$, $a' \in A'$, being two assertions. Then

$$T \cup A \models a \quad \text{iff} \quad A' \models a' \quad \text{iff} \quad a' \in A'$$

Theorem evidence (LoDE entity reasoning). The theorem above is a corollary of the main result of expansion, see above.

LoDE entity reasoning

Observation (LoDE entity reasoning). The theorem above is the main result which also summarizes how expansion works. It can be summarized as follows:

- T is a LoD teleontology
- A is an LoE EG
- T is unfolded in T' and then T' is exploited to expand A into A'
- A' is an extended EG where the original nodes and links of the EG corresponding to A have been extended with the conjuncts resulting from the expansion of A with respect to T'
- Once you have A', you can apply LoE reasoning to $T = A'$

LoDE Entity Reasoning - example

$T = \{ \text{FemaleMother} \equiv \text{Female} \sqcap \forall \text{HasChild.female}, \text{Female} \equiv \text{Person} \sqcap \text{Nice} \}$

$A = \{ \text{FemaleMother}(\text{Anna}), \text{HasChild}(\text{Anna}, \text{Mary}) \}$

$C = \text{FemaleMother}$

The expansion A' of A with respect to T is:

$A' = \{ \text{FemaleMother}(\text{Anna}), \text{HasChild}(\text{Anna}, \text{Mary}), \text{Female}(\text{Anna}), \text{Female}(\text{Mary}), \text{Person}(\text{Anna}), \text{Nice}(\text{Anna}), \text{Person}(\text{Mary}), \text{Nice}(\text{Mary}) \}$

$\models \text{Person}(\text{Anna})?$

SI!

$\models \text{FemaleMother}(\text{Mary})?$

IDK if OWA! No if CWA!

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Key notions

- Knowledge Base, expert system
- Entity Base
- Etype unfolding
- TBOX unfolding
- Etype expansion



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